

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2, BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Annexure A

18
20
20
Total Marks: 25

Comparative Analysis

Code of Conduct: (192072128)

M. Modhulika (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Modhulika

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Concepts	15%	Demonstrates complete and in-depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

1. Compare Star and Bus Topology in terms of Cost, Reliability and Performance

Star topology connects every device to a central device such as a switch or hub. **Bus topology** connects all devices to a single backbone cable.

Cost

- Bus topology is less expensive because it uses a single backbone cable.
- Star topology requires individual cables from every device to the central switch.
- Star also requires a switch or hub, increasing the initial cost.
- Therefore, bus is more economical for a small network.

Reliability

- Star topology is more reliable because failure of one cable or computer normally affects only that device.
- In bus topology, failure of the main backbone cable can affect the complete network.
- The central switch is an important point in star topology, so its failure can affect all connected devices.

Performance

- Star topology generally provides better performance.
- A switch can control and forward data efficiently to the required device.
- In bus topology, all devices share the same communication medium.
- As more devices are connected, network traffic and collisions can increase.

Example

A school computer laboratory can use star topology because each computer can be connected individually to a central switch.

Conclusion

Bus topology is better for low-cost and simple networks, while star topology is better for performance, reliability and easy management.

2. Compare Mesh and Star Topology in terms of Fault Tolerance

Mesh topology provides multiple connections between devices, whereas star topology uses a central switch or hub.

Mesh Topology

- Each device has connections to multiple other devices.
- Multiple paths are available for data transmission.
- If one connection fails, another path can be used.
- It provides very high fault tolerance.
- It is suitable for critical communication systems.

Star Topology

- All devices are connected to a central switch or hub.
- If one computer fails, the other computers can continue working.
- However, if the central switch fails, communication between connected devices can be interrupted.
- Therefore, star has lower fault tolerance than mesh.

Comparison

Parameter	Mesh Topology	Star Topology
Fault tolerance	Very high	Moderate
Alternative paths	Multiple	Limited
Central device	Not required	Required
Reliability	Very high	High
Cabling	Very high	Moderate
Cost	High	Lower than mesh
Maintenance	Difficult	Easy

Example

Mesh topology can be useful in critical networks where continuous communication is important, while star topology is commonly suitable for offices and laboratories.

Conclusion

Mesh topology is more fault tolerant than star topology because it provides multiple alternative communication paths.

3. Compare Bus and Mesh Topology in terms of Installation Complexity

Bus Topology

Bus topology uses a single backbone cable to which multiple devices are connected.

Advantages during installation:

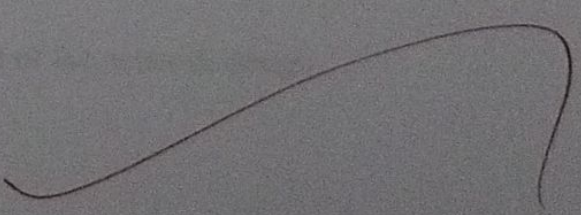
- Simple network structure.
- Requires less cable.
- Requires fewer connections.
- Easy to install for small networks.
- Low installation cost.

However, finding faults in the backbone can be difficult because one cable is shared by many devices.

Mesh Topology

Mesh topology provides connections between multiple devices.

Disadvantages during installation:

- Requires a large number of cables.
 - Requires many ports and connections.
 - Installation takes more time.
 - Network design is more complicated.
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- Cost increases as more devices are added.

For n devices, a full mesh requires:

$$\text{Number of links} = n(n - 1) / 2$$

For example, if there are 5 devices:

$$5(5 - 1) / 2 = 10 \text{ links}$$

So, the number of connections increases rapidly as devices are added.

Comparison

Parameter	Bus	Mesh
Installation	Easy	Difficult
Cabling	Less	More
Cost	Low	High
Complexity	Low	High
Expansion	Limited	Complex
Troubleshooting	Backbone faults can be difficult	Many connections make it difficult

Conclusion

Bus topology is simpler, cheaper and faster to install, whereas mesh topology provides greater reliability but requires complex and costly installation.

4. Compare Hybrid and Star Topology in terms of Scalability

Scalability refers to how easily a network can be expanded when more devices or network segments are added.

Hybrid Topology

Hybrid topology combines two or more different topologies such as:

- Star
- Bus
- Mesh
- Ring

This allows organizations to design a network according to their requirements.

Advantages of Hybrid Topology

- Highly flexible.
- Can support large networks.
- Different network sections can use different topologies.
- New sections can be added without completely changing the existing network.
- Suitable for large organizations.

Star Topology

Star topology can also be expanded by adding more devices to the central switch.

However:

- Expansion depends on available switch ports.
- If all ports are occupied, an additional switch may be required.
- The central device becomes an important part of the network design.

Comparison

Parameter	Hybrid	Star
Scalability	Very high	Good
Flexibility	Very high	Moderate
Expansion	Highly flexible	Depends on central device
Network size	Large	Small to medium
Complexity	High	Low
Cost	Higher	Lower

Example

A large university campus may use a hybrid topology by using star topology in individual departments and connecting different departments using another topology.

Conclusion

Hybrid topology is more scalable and flexible than star topology because different network structures can be combined according to future requirements.

5. Compare Star, Mesh, Bus and Hybrid Topologies in terms of Maintenance

Maintenance involves identifying faults, repairing connections, monitoring devices and keeping the network operational.

Star Topology

- Easy to monitor.
- Individual cable failures can be isolated.
- Faulty computers can be identified easily.
- Central switch simplifies network management.
- Therefore, maintenance is relatively easy.

Mesh Topology

- Highly reliable but difficult to maintain.
- Contains many connections.
- Troubleshooting requires checking multiple links.
- Repairs can take more time.
- Requires skilled network administrators.

Bus Topology

- Simple structure but maintenance can be difficult.
- A backbone cable is shared by all devices.
- A fault in the backbone can affect the entire network.
- Locating the exact fault can be difficult.

Hybrid Topology

- Maintenance complexity is moderate to high.
- Different sections may use different topologies.
- Administrators must understand each topology.
- Troubleshooting can vary between network segments.

- Skilled administration may be required.

Comparison

Topology	Maintenance	Main Reason
Star	Easy	Faults are easy to isolate
Mesh	Difficult	Many connections
Bus	Difficult	Backbone fault affects network
Hybrid	Moderate	Multiple topologies are combined

Conclusion

Star topology is the easiest to maintain. Mesh provides high reliability but requires more maintenance effort. Bus is simple but backbone failures are difficult to troubleshoot, while hybrid requires careful management because it combines different topologies.

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